In [1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris

iris = load_iris()

data = pd.DataFrame(iris.data, columns=[
    'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm'
])

data['Id'] = range(1, 151)
data['variety'] = ['Iris-' + iris.target_names[i] for i in iris.target]

data = data[['Id', 'SepalLengthCm', 'SepalWidthCm',
             'PetalLengthCm', 'PetalWidthCm', 'variety']]

data

data.info()

data.describe()

print(data.value_counts('variety'))

sns.countplot(x='variety', data=data)
plt.show()

dummies = pd.get_dummies(data.variety)

FinalDataset = pd.concat(
    [pd.get_dummies(data.variety), data.iloc[:, [0, 1, 2, 3]]],
    axis=1
)

print(FinalDataset.head())

sns.scatterplot(x='SepalLengthCm', y='SepalWidthCm',
                hue='variety', data=data)
plt.show()

sns.scatterplot(x='PetalLengthCm', y='PetalWidthCm',
                hue='variety', data=data)
plt.show()

sns.pairplot(data, hue='variety', height=3)
plt.show()

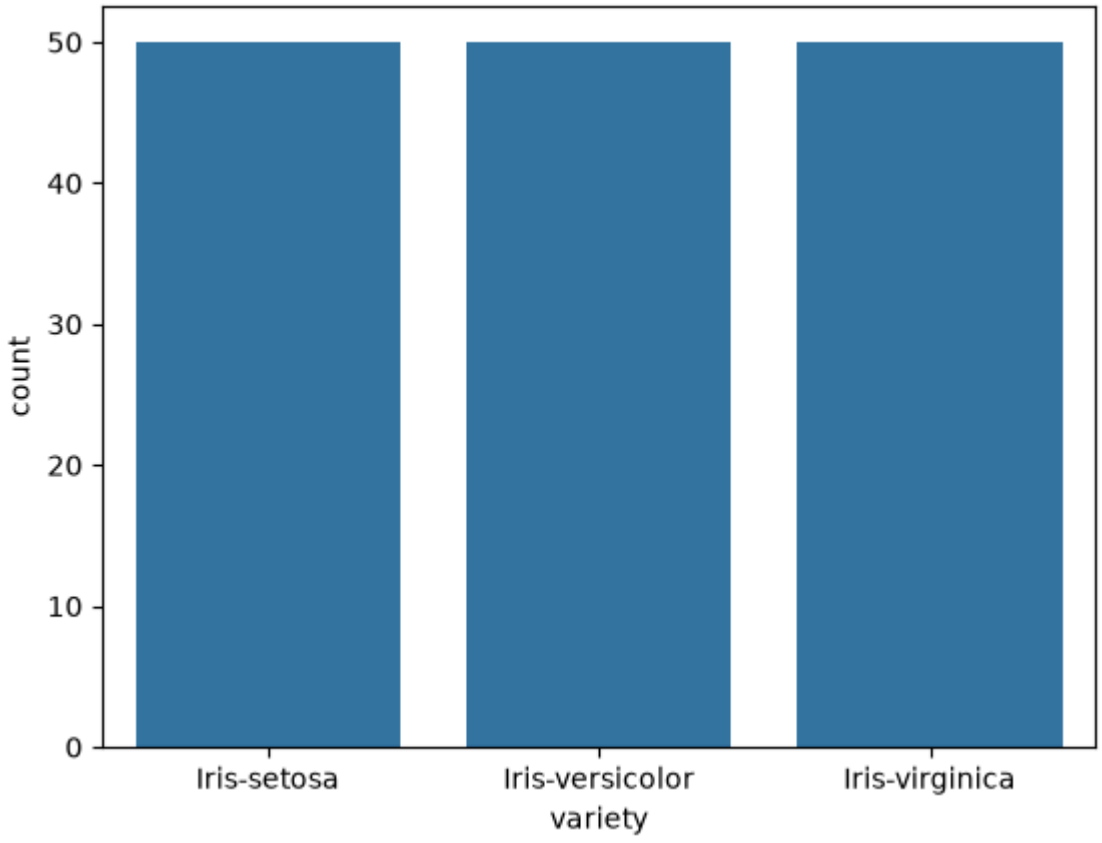
sns.FacetGrid(data, hue='variety', height=3).map(
    sns.histplot, 'PetalLengthCm'
).add_legend()
plt.show()

sns.FacetGrid(data, hue='variety', height=3).map(
    sns.histplot, 'PetalWidthCm'
).add_legend()
plt.show()
```

```
sns.FacetGrid(data, hue='variety', height=3).map(
    sns.histplot, 'SepalLengthCm'
).add_legend()
plt.show()

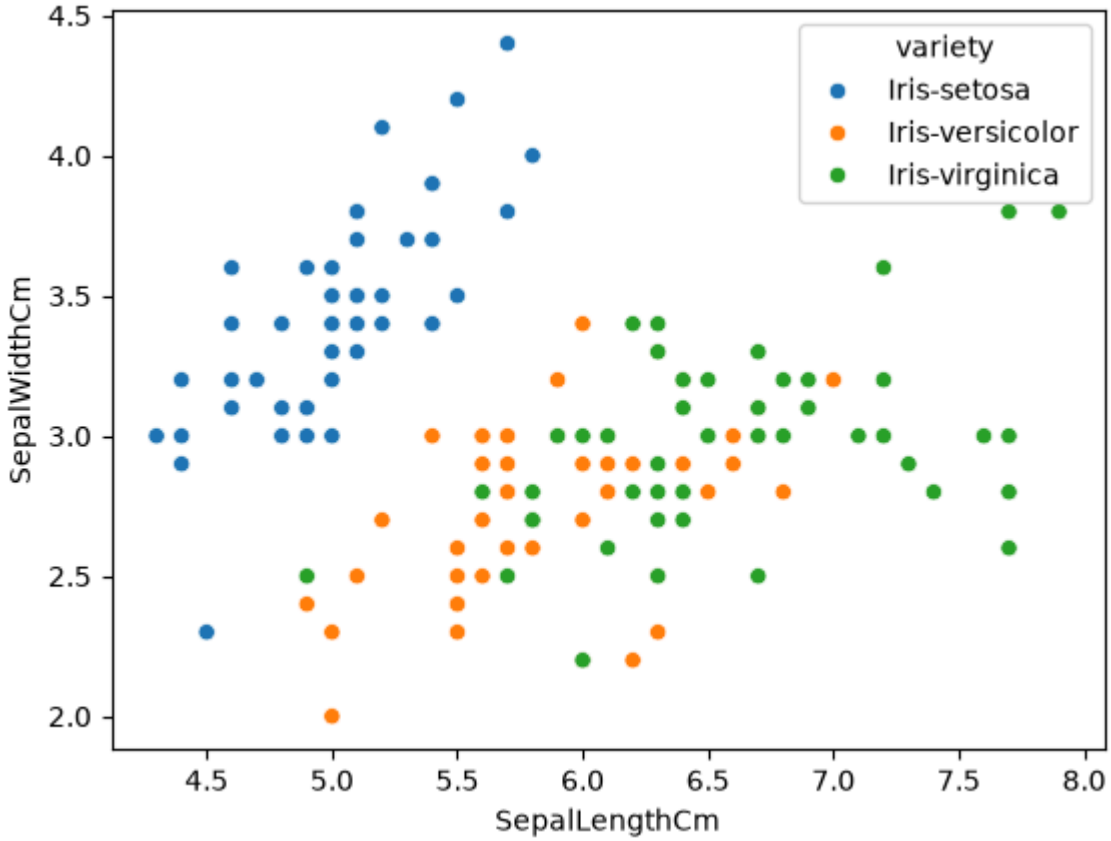
sns.FacetGrid(data, hue='variety', height=3).map(
    sns.histplot, 'SepalWidthCm'
).add_legend()
plt.show()
```

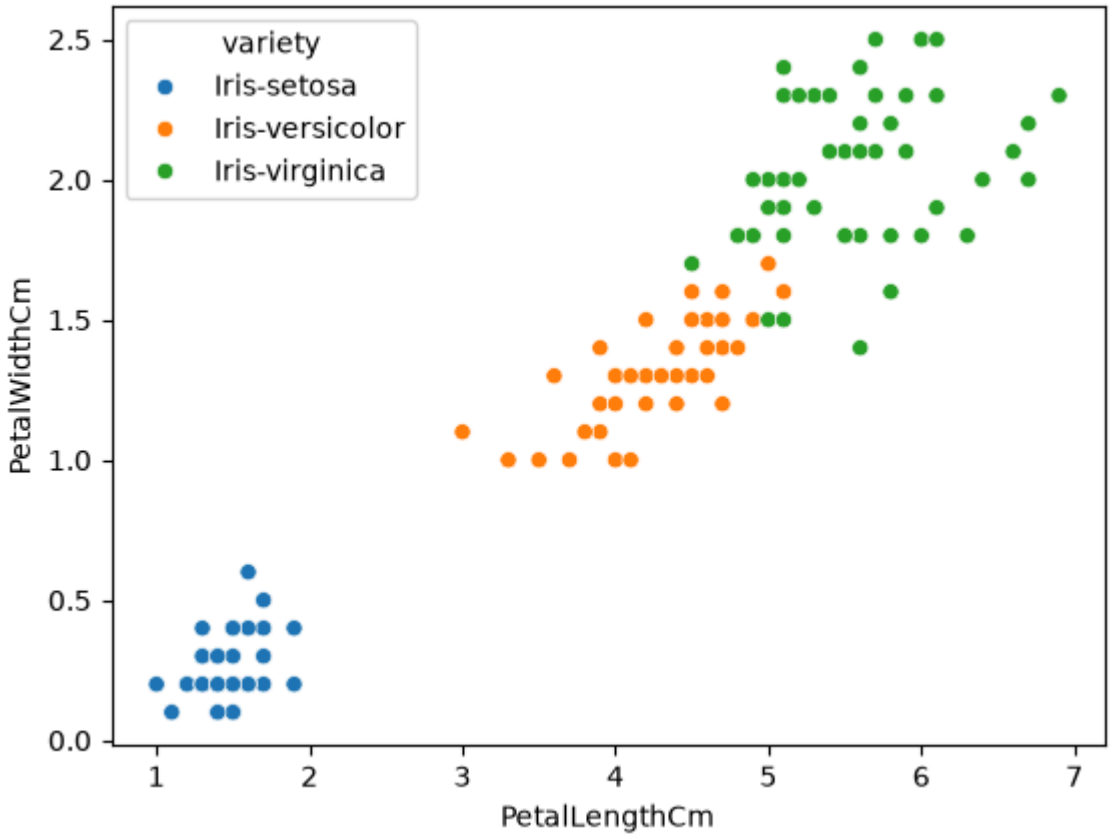
```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   variety        150 non-null    str
dtypes: float64(4), int64(1), str(1)
memory usage: 9.1 KB
variety
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```

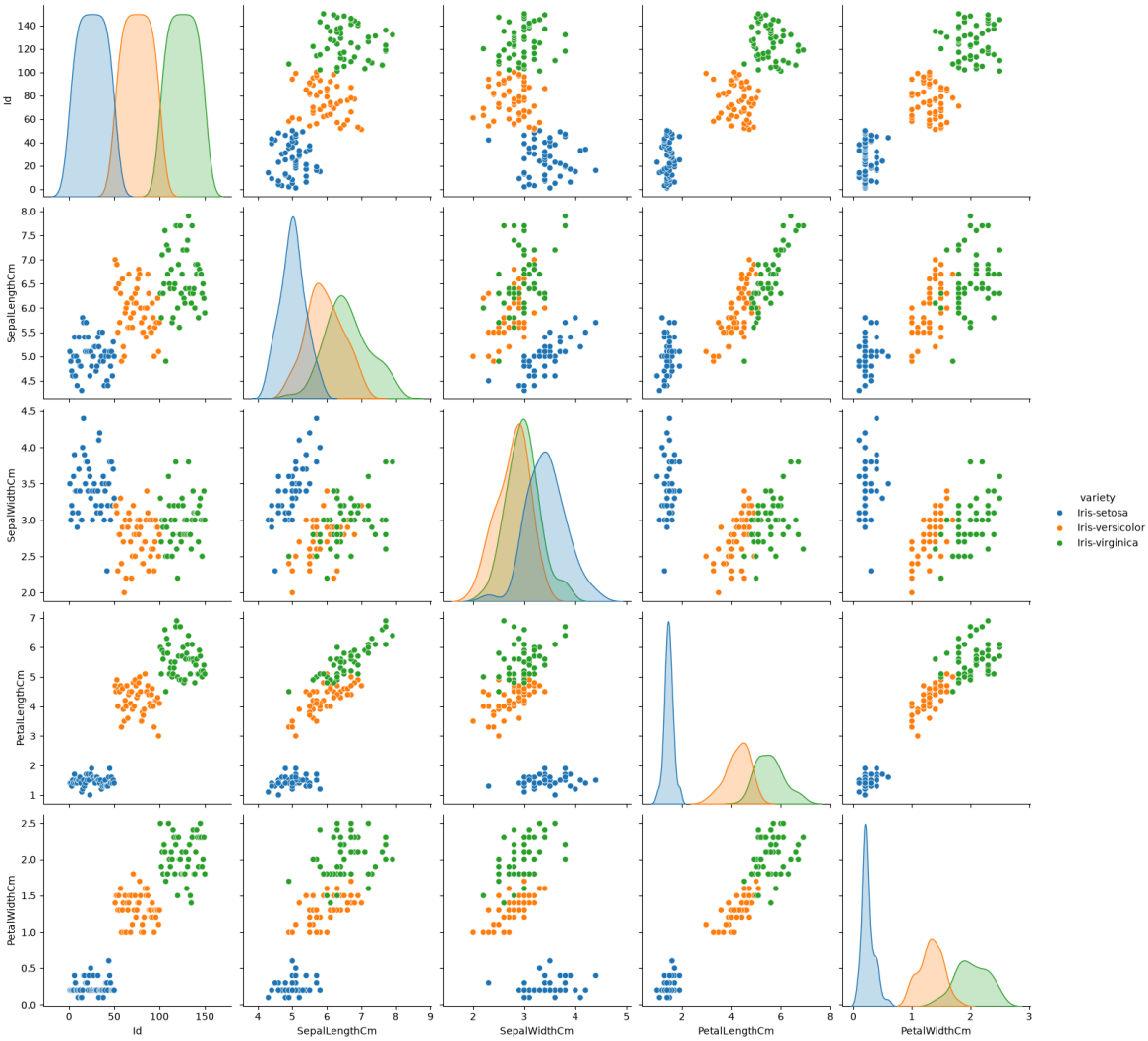


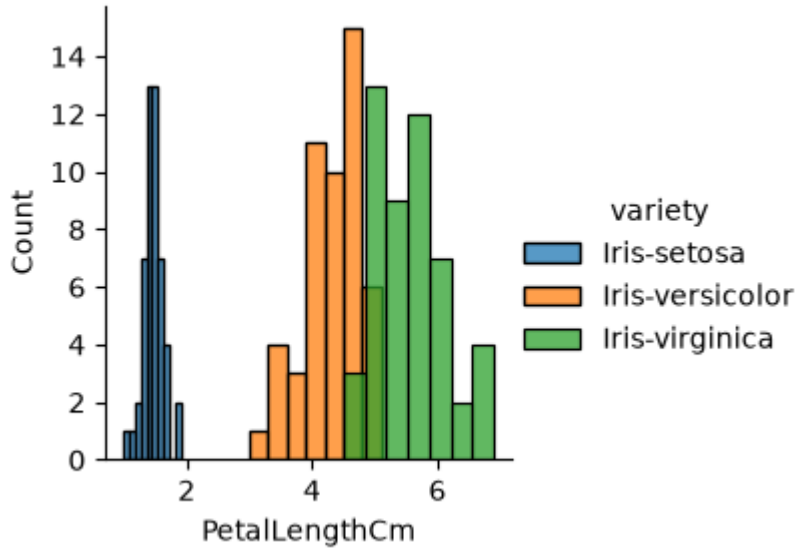
```
   Iris-setosa  Iris-versicolor  Iris-virginica  Id  SepalLengthCm  \
0         True            False           False   1            5.1
1         True            False           False   2            4.9
2         True            False           False   3            4.7
3         True            False           False   4            4.6
4         True            False           False   5            5.0

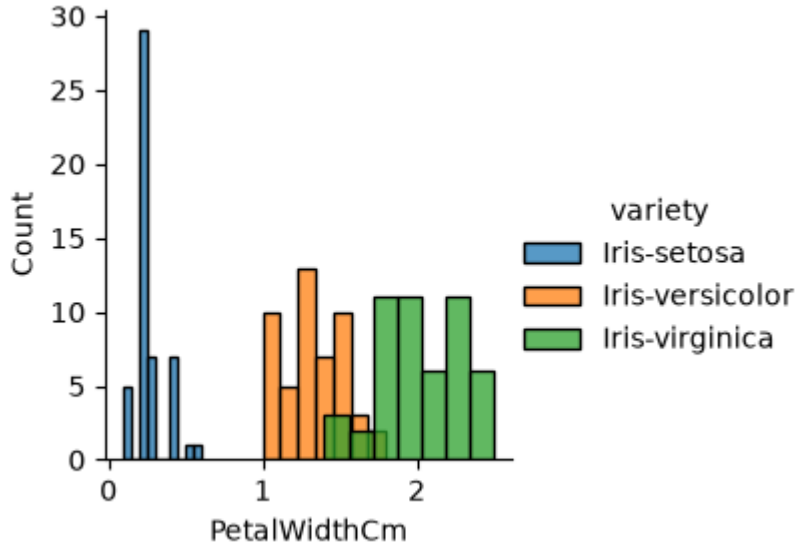
   SepalWidthCm  PetalLengthCm
0           3.5            1.4
1           3.0            1.4
2           3.2            1.3
3           3.1            1.5
4           3.6            1.4
```

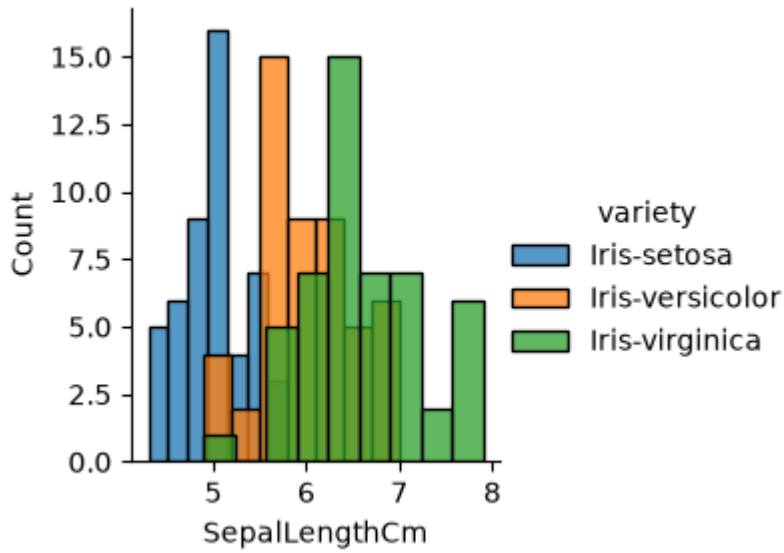


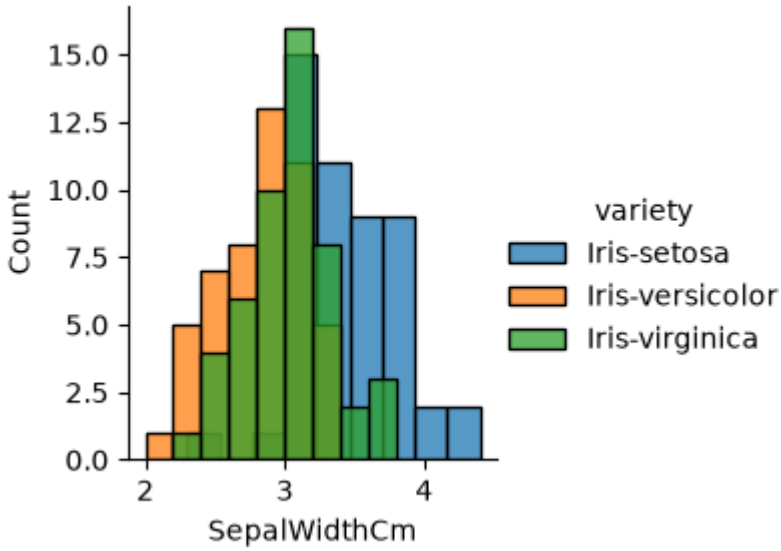

variety
Iris-setosa
Iris-versicolor
Iris-virginica
PetalWidthCm
PetalLengthCm


Id
SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
SepalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


15.0
12.5
10.0
7.5
5.0
2.5
0.0
Count
2
3
4
SepalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica

In [ ]: